SHRIHARI M V

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Professional Summary

Computer Science (Data Science) graduate focused on **Generative AI and AI engineering**, with hands-on experience building end-to-end **RAG and LLM applications** across data preparation, retrieval, reranking, context construction, generation, evaluation, and API integration. Built and tested **tool-using, stateful LLM workflows** that connect model decisions with backend data and services, with emphasis on retrieval quality, grounded responses, and reliable system behavior.

Technical Skills

Programming	Python, SQL, Java, TypeScript
Machine Learning	Scikit-learn, XGBoost, Feature Engineering, Cross-Validation, Model Evaluation, Calibration
Deep Learning	PyTorch, TensorFlow, Transformers, Swin Transformer, Transfer Learning, OpenCV, Explainable AI
Generative AI	LLMs, Prompt Engineering, Structured Outputs, RAG, Embeddings, Semantic Search, Vector Databases
Agentic AI	LangChain, LangGraph, Tool Calling, Stateful Workflows, Conditional Routing, Agent Memory, Human-in-the-Loop
AI Engineering	FastAPI, REST APIs, AsyncIO, Retrieval Evaluation, Experimentation, Error Analysis, Structured Logging, Git/GitHub
Data / Deployment	Pandas, NumPy, ChromaDB, Ollama, MLflow, Docker, ONNX Runtime, Streamlit, Gradio

Projects

StudyMentor – Adaptive Agentic Learning Platform GitHub

Python • FastAPI • LangChain • LangGraph • Ollama • Pydantic • Pytest

- Built an adaptive learning workflow that transforms learner retention, review history and concept gaps into structured state, enabling an LLM-based planner to select targeted **Retrieval, Feynman or Elaboration** activities.
- Integrated **LLM tool calling** with Pydantic-validated outputs, connecting model decisions to backend tools for weak-topic analysis and spaced-repetition scheduling.
- Orchestrated the workflow with **LangGraph** using typed state, conditional routing, checkpointing, bounded retries and human-review escalation, with asynchronous learner-memory loading.
- Validated agent behavior using deterministic quality checks and integration tests; achieved **19/19 passing backend tests** and demonstrated memory-driven selection of a weak topic for targeted learning.

CarnaticPrep – Retrieval-Augmented Knowledge System GitHub

Python • ChromaDB • Sentence Transformers • CrossEncoder • Ollama • FastAPI • RAGAS

- Designed a modular backend implementing hybrid **Retrieval-Augmented Generation** with dense retrieval, **BM25**, weighted Reciprocal Rank Fusion (RRF), and **CrossEncoder reranking**.
- Built knowledge corpus using **Wikipedia APIs, BeautifulSoup, PDF and CSV processing**, with cleaning, metadata normalization, structure-aware chunking, and vector indexing.
- Developed configurable **FastAPI** workflows for document ingestion, retrieval, context construction, and grounded response generation, with structured logging and OpenAPI documentation.
- Experimenting with **chunking, retrieval weighting, reranking, and context construction** strategies; evaluating retrieval and generation quality using **RAGAS** on a benchmark query set.

AksharaVision – Kannada Handwritten Character Recognition System GitHub

Python • PyTorch • Swin Transformer • OpenCV • FastAPI • Gradio • TinyDB • ONNX Runtime

- Curated a balanced **56.5K-image, 113-class** Kannada OCR dataset using preprocessing, augmentation and image-quality validation pipelines.
- Benchmarked CNN and Vision Transformer architectures and selected **Swin Transformer** through controlled experimentation on model performance and representation quality.
- Improved prediction reliability with **temperature scaling**, achieving **Macro-F1 0.9989, ECE 0.00065 and Brier Score 0.00112**, supported by systematic failure analysis.
- Built **FastAPI inference workflows** with Top-3 predictions, telemetry, feedback collection, PSI-based drift monitoring and **ONNX Runtime** deployment support.

Education

B.E. Computer Science & Engineering (Data Science) 2022 – 2026
Bangalore Institute of Technology CGPA: **8.6/10**

PCMB 2020 – 2022
Sri Kumarans PU College Percentage: **94.83%**

Experience

MindMatrixEd – Generative AI Intern Feb 2026 – May 2026

- Explored AI-assisted Android application workflows using Google AI Studio, Android Studio and Kotlin contributing to feature prototyping and user-centric product design.